ADJUNCTION OF PULLBACK AND PUSHFORWARD FOR STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Reduction-type proof; the cited lemma-stackify-universal-property-more carries an 'Omitted' proof in the source and lemma-adjointness-pullback-pushforward omits its quasi-inverse verification; both are used as declared references.

Allowed standard prerequisites (editorial).

- Adjunction for fibred categories. lemma-adjointness-pullback-pushforward (same chapter, named). Provides the fibred-level adjunction equivalence Mor_{Fib/C}(S, u^pT) = Mor_{Fib/D}(u_pS, T).
- Universal property of stackification. lemma-stackify-universal-property-more (same chapter, named). Extends the adjunction across stackification: Mor_{Fib/C}(S, X) = Mor_{Stacks/C}(S', X) with S' the stackification.
- Stacks versus fibred categories. definition-stacks-over-C (same chapter, named). Identifies i-morphisms of stacks with i-morphisms of fibred categories for i = 1, 2.

Source reference. Original definition 02ZG: sources/stacks-stacks/pinned-original-source.tex, source lines 917–917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04WI: sources/stacks-stacks/pinned-original-source.tex, source lines 3384–3384. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04W9: sources/stacks-stacks/pinned-original-source.tex, source lines 1988–1988. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00X6: focused-seeds/geometry/S9-stk-adjointness-pullback-pushforward-stacks/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{D} \to \mathcal{C}$ be a morphism of sites given by a continuous functor $u : \mathcal{C} \to \mathcal{D}$ satisfying the hypotheses and conclusions of Sites, Proposition 00X6. Let $p : \mathcal{S} \to \mathcal{C}$ and $q : \mathcal{T} \to \mathcal{D}$ be stacks. Then we have a canonical equivalence of categories*

$$\mathrm{Mor}_{Stacks/\mathcal{C}}(\mathcal{S}, f_*\mathcal{T}) = \mathrm{Mor}_{Stacks/\mathcal{D}}(f^{-1}\mathcal{S}, \mathcal{T})$$

of morphism categories.

Proof. For $i = 1, 2$ an i-morphism of stacks is the same thing as a i-morphism of fibred categories, see Definition 02ZG. By Lemma 04WI we have already

$$\mathrm{Mor}_{Fib/\mathcal{C}}(\mathcal{S}, u^p\mathcal{T}) = \mathrm{Mor}_{Fib/\mathcal{D}}(u_p\mathcal{S}, \mathcal{T})$$

Hence the result follows from Lemma 04W9 as $u^p\mathcal{T} = f_*\mathcal{T}$ and $f^{-1}\mathcal{S}$ is the stackification of $u_p\mathcal{S}$. □